

Strategic coexistence theory for evolutionary games

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Abstract

Evolutionary game theory and ecological coexistence theory both seek to predict the outcome of competition between biological entities, be they strategies or species, but the two fields have relied on largely separate approaches. Replicator equations provide a foundation for analyzing strategy competition and the stability of strategic equilibria, yet they do not explicitly separate the mechanisms that stabilize competition from those that equalize fitness differences between strategies. Here, we extend modern coexistence theory from community ecology to develop strategic coexistence theory (SCT), a framework for quantifying strategic niche and fitness differences between competing strategies. SCT recovers the classic classification of two-strategy games, distinguishing competitive exclusion, coexistence, and priority effects within a shared niche–fitness difference space. Applying SCT to five mechanisms for the evolution of cooperation further reveals that these mechanisms promote cooperation through distinct dynamical routes: kin selection, network reciprocity, and group selection primarily reduce fitness differences, whereas direct and indirect reciprocity destabilize competition and generate priority effects. Finally, applying SCT to microbial public-goods game shows that nonlinear microbial growth can both stabilize and equalize competition between cooperators and defectors, allowing coexistence. Together, these results show that SCT provides a complementary framework for comparing evolutionary games and teasing apart the coexistence mechanisms underlying strategy competition.

Introduction

Understanding the mechanisms that promote the evolutionary stability of competing strategies is a fundamental aim in evolutionary biology (Lewontin, 1961; Slobodkin, 1964; Smith and Price, 1973; Taylor and Jonker, 1978). To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition through payoff inequalities, invasion criteria, and stability analysis (Schuster and Sigmund, 1983; Hofbauer and Sigmund, 1998; Cressman, 2003; Shalizi, 2009; Cressman and Tao, 2014). However, less attention has been given to similarities between replicator dynamics and community ecology until recently (Gokhale and Traulsen, 2025; Tarnita and Traulsen,

2025; Allesina, 2026), even though the classic replicator equations and the generalized Lotka-Volterra equation are mathematically equivalent (Hofbauer, 1981; Page and Nowak, 2002).

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025): the dominance of a single strategy (prisoner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a single strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory (Motro, 1991; Doebeli et al., 2004; Hauert and Doebeli, 2004; Doebeli and Hauert, 2005; Gore et al., 2009; Souza et al., 2009; Archetti and Scheuring, 2011; Steidinger and Bever, 2014; Tarnita and Traulsen, 2025). In fact, stable polymorphism of evolutionary strategies, such as coexistence of cooperators and defectors, are observed in microbial and other biological systems (Gore et al., 2009; Cordero et al., 2012; Kocher et al., 2018; Pollak et al., 2021; Lin et al., 2024), but coexistence-theoretic perspective has often remained implicit. Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition (Chesson, 2000)? While EGT allows us to predict whether two strategies can coexist or not, a framework for quantifying coexistence mechanisms between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for quantifying coexistence mechanisms (Chesson, 2000; Godoy et al., 2014; Godoy and Levine, 2014; Kraft et al., 2015; Adler et al., 2026). MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to overcome average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor (Chesson, 2000). In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions. Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each entity to compete under the limiting environments created by its competitor (Chesson, 2000). This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains (Sieben

et al., 2022; Park et al., 2024, 2026). Inspired by this effort, we present a Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two-strategy games, providing reinterpretation of classic results from coexistence-theoretic perspective. We then apply SCT to comparing five mechanisms for the evolution of cooperation, revealing that direct and indirect reciprocity destabilize competition between cooperation and defection, thus pushing the system towards priority effect regime. Finally, we consider a case study of microbial cooperation, where nonlinear microbial growth can both stabilize and equalize competition between cooperation and defection, allowing coexistence of strategies.

Strategic coexistence theory

Here, we present a brief derivation of strategic coexistence theory (SCT) (Figure 1). To do so, we begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs (Schuster and Sigmund, 1983; Hofbauer and Sigmund, 1998; Cressman, 2003; Shalizi, 2009; Cressman and Tao, 2014):

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j .

Because payoffs in many games can be negative, payoff ratios can be difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

This exponential transformation is convenient because it preserves the payoff ordering and does not alter the invasion conditions. Therefore, this allows us to compare strategic environments through ratios, as in modern coexistence theory. For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of

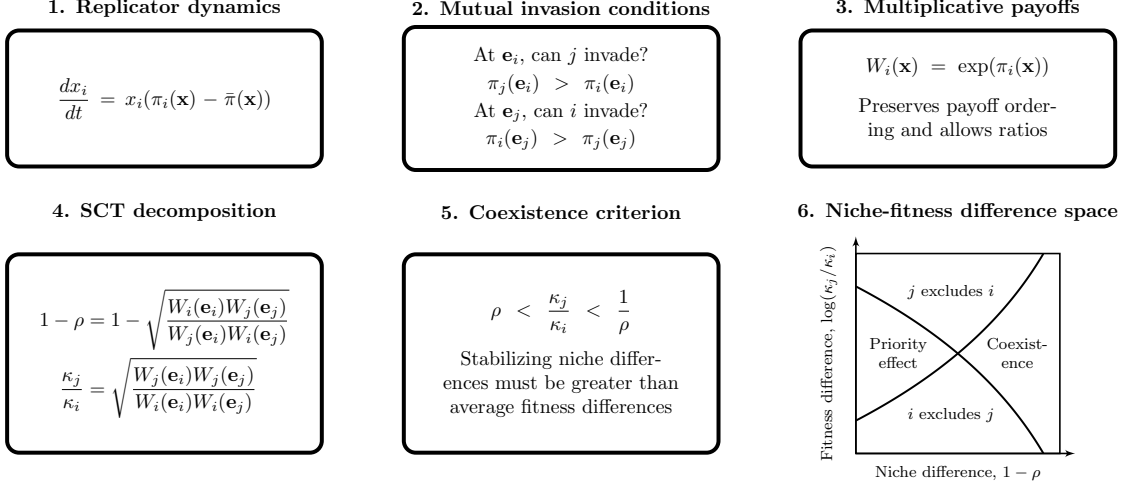


Figure 1: **A schematic diagram summarizing strategic coexistence theory.** Strategic coexistence theory decomposes pairwise replicator dynamics into stabilizing and equalizing mechanisms. Mutual invasion conditions can be rewritten on a multiplicative payoff scale, allowing us to derive strategic niche difference $1 - \rho$ and fitness difference κ_j/κ_i . Then, the inequality $\rho < \kappa_j/\kappa_i < 1/\rho$ determines mutual invasion.

each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

When $1 - \rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition. Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

In other words, strategy j has an average competitive advantage over strategy i ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated by itself and its competitor.

Together, for any two strategies described by the classic replicator equation, mutual invasion requires stabilizing niche differences to overcome average fitness differences. Specifically, the condition for mutual invasion can be rewritten as (Chesson,

129 2000):

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

130 This inequality applies to any games described by the classic replicator equation
 131 because ρ and κ_j/κ_i can be calculated directly from ratios of multiplicative pay-
 132 offs. This decomposition not only allows us to understand how different mechanisms
 133 contribute to the coexistence or exclusion of competing strategies but also provides
 134 shared axes for comparing different games. Detailed derivations are provided in
 135 Supplementary Text S1.

136 We note that the absolute magnitudes of niche and fitness differences are sensitive
 137 to payoff scaling because they are defined on an exponentiated payoff scale. However,
 138 the qualitative outcome of strategic competition under SCT is preserved because the
 139 exponential transformation does not alter invasion conditions.

140 Classic two-strategy games

141 As an illustrative example, we begin with classic two-strategy games (Hauert, 2002;
 142 Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025), which can be
 143 described by the following payoff matrix (Figure 2A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

144 Here, the first and second rows represent strategies 1 and 2, respectively, and the
 145 columns represent the strategy of the interacting opponent. Writing x_1 to denote
 146 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

147 Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

148 The signs of $T - R$ and $S - P$ determine whether each strategy can invade the
 149 other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas
 150 strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition
 151 the game space into four regimes (Figure 2B): prisoner’s dilemma ($T > R > P >$
 152 S), hawk–dove game ($T > R, S > P$), stag–hunt game ($R > T, P > S$), and
 153 harmony game ($R > T, S > P$). In standard EGT, these outcomes are obtained by
 154 analyzing the stability of equilibrium conditions. For simple games, this analysis is
 155 straightforward.

156 A coexistence-theoretic approach recovers the same classification but provides
 157 a different interpretation (Figure 2C). For a two-strategy game determined by this

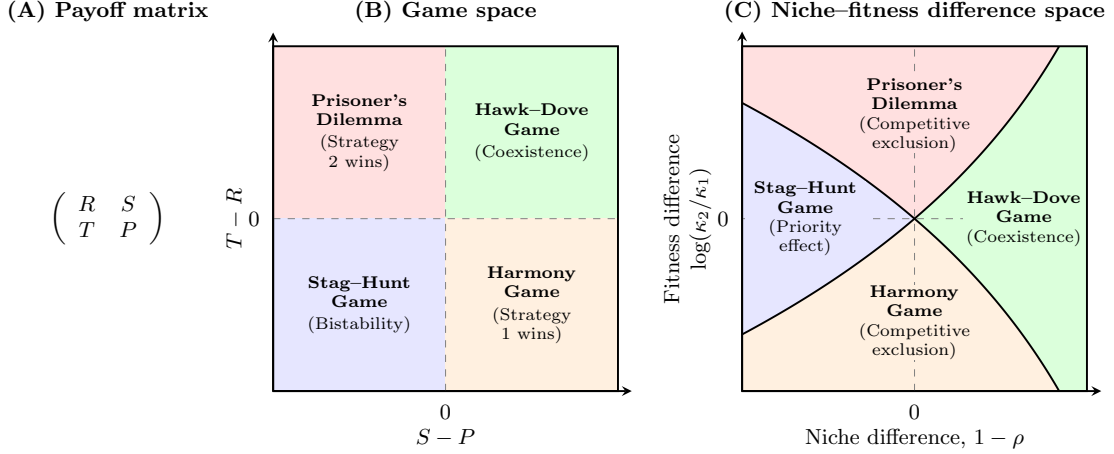


Figure 2: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk-dove games allow coexistence; and stag-hunt games result in bistability. (C) Mapping of the same four games onto niche-fitness difference space. Hawk-dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag-hunt games correspond to the priority-effect regime. $R > P$ is assumed throughout, but this does not affect coexistence outcome in any way.

158 payoff matrix, the niche difference and fitness difference are given by the following
 159 set of equations (Supplementary Text S2):

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

160 As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to
 161 $T > R$ and $S > P$. This corresponds to the hawk-dove game, in which each strategy
 162 can invade its competitor when rare and the two strategies coexist. The remaining
 163 games also have natural interpretations in niche-fitness difference space. Prisoner's
 164 dilemma and harmony games represent competitive exclusion regimes, where one
 165 strategy has a fitness advantage that exceeds niche differences. In contrast, stag-hunt
 166 games represent a priority-effect regime, where neither strategy can invade the other
 167 and the outcome depends on initial conditions. Therefore, coexistence theory not
 168 only allows us to recover the classic results from two-strategy games but also provides

new interpretations for these results in terms of niche and fitness differences. Building on this, we next apply this framework to comparing core mechanisms of cooperation.

Comparisons of mechanisms for the evolution of cooperation

Even though cooperative behavior can be found across a wide range of biological systems, a simple prisoner’s dilemma shows that cooperation is not an evolutionarily stable strategy (May, 1987). Thus, understanding mechanisms that promote the evolution of cooperation has remained a central goal in evolutionary biology (West et al., 2007). Here, we compare five classic mechanisms for the evolution of cooperation using SCT (Nowak, 2006) and ask how they differ from the perspective of SCT.

Specifically, we consider the following mechanisms: kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection (Figure 3A). Kin selection favors cooperation when two interacting individuals are genetic relatives, where the genetic relatedness r must be greater than the cost-to-benefit ratio: $r > c/b$ (Hamilton, 1964). Direct reciprocity favors cooperation when current cooperation may promote future cooperation by the same individual through repeated encounters (Trivers, 1971; Axelrod and Hamilton, 1981). In this case, the probability of repeated interaction (i.e., “next round”) must be greater than the cost-to-benefit ratio: $w > c/b$. Indirect reciprocity favors cooperation when current cooperation may promote future cooperation by different individuals through reputation (Nowak and Sigmund, 1998; Wedekind and Milinski, 2000; Nowak and Sigmund, 2005). Thus, the social acquaintanceship (i.e., the probability of knowing someone’s reputation) must be greater than the cost-to-benefit ratio: $q > c/b$. Network reciprocity favors cooperation via clustering of cooperators, requiring the benefit-to-cost ratio to be greater than the average number of neighbors: $b/c > k$ (Nowak and May, 1992; Lieberman et al., 2005). Finally, group selection favors cooperation via multilevel selection, where groups with more cooperators exhibit faster growth at the group level even if defectors reproduce faster at the individual level (Traulsen and Nowak, 2006). This requires $b/c > 1 + (n/m)$, where n is the maximum group size and m is the number of groups. All of these mechanisms can be described by a simple 2×2 payoff matrix (Nowak, 2006), from which the corresponding niche and fitness differences can be computed using Eq. (10) and Eq. (11).

First, the prisoner’s dilemma provides a baseline scenario, where defection is evolutionarily stable (Figure 3A). From the perspective of SCT, there is complete niche overlap, and therefore defection competitively excludes cooperation (Figure 3B). Building on this, SCT shows that five mechanisms promote evolutionary stability of cooperation via different dynamical routes (Figure 3B). For example, kin selection allows the fitness of cooperation to increase as genetic relatedness r increases (Figure 3B). However, it does not affect the niche difference between cooperation and

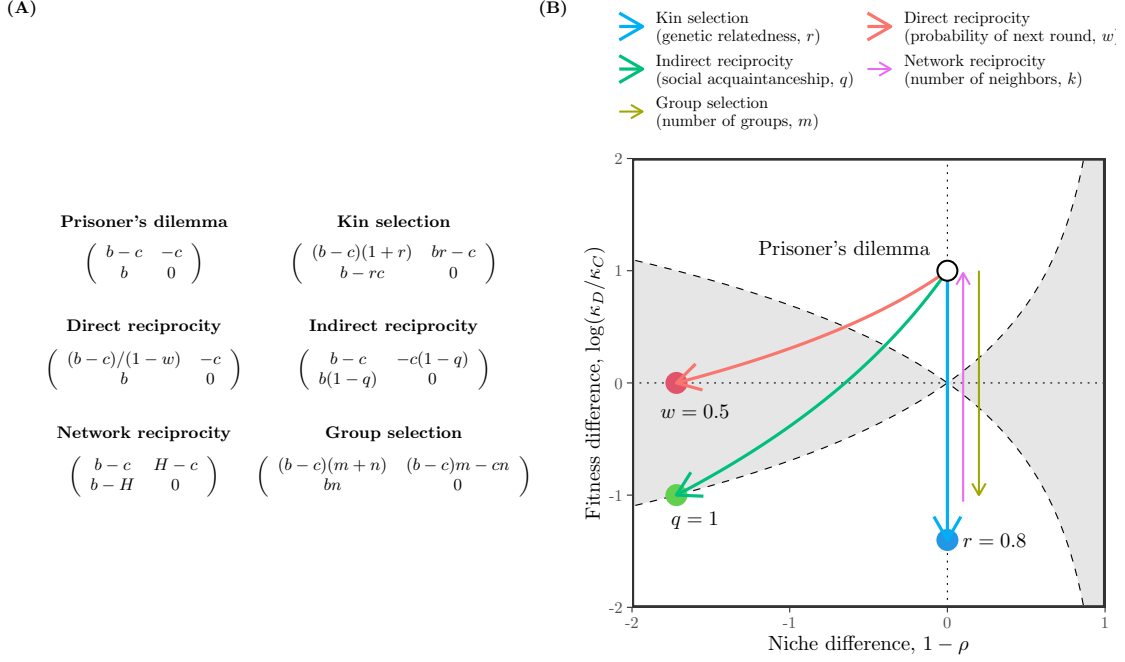


Figure 3: **Comparisons of five mechanisms for the evolution of cooperation using SCT.** (A) Payoff matrices for the baseline scenario (prisoner's dilemma) and five mechanisms for the evolution of cooperation (Nowak, 2006). Prisoner's dilemma is characterized by the benefit for the recipient b and the cost for the donor c . Network reciprocity is parameterized based on $H = [(b-c)k - 2c]/[(k+1)(k-2)]$. (B) Changes in niche and fitness differences driven by each mechanism. Each arrow represents the direction of change as we increase the corresponding parameter, indicated in figure legends. Note that kin selection, network reciprocity, and group selection all result in changes in fitness difference without causing any changes in niche difference. Therefore, we indicate the direction of change for network reciprocity and group selection separately to avoid overlapping arrows.

defectors. Similarly, network reciprocity and group selection modulate the fitness of cooperation without causing any changes in strategic niches. In contrast, direct and indirect reciprocity destabilize competition while increasing the fitness of cooperation, thereby pushing the competition outcome towards a priority effect regime. Equivalently, these two mechanisms promote bistability of defection and cooperation. SCT further reveals another difference between direct and indirect reciprocity. In particular, direct reciprocity is unable to generate sufficient fitness difference to competitively exclude defection even when the probability of next round w reaches 1. In contrast, indirect reciprocity is able to generate sufficient fitness difference to push the system towards the boundary between priority-effect regime and competitive-exclusion regime as social acquaintanceship q reaches 1, indicating that an additional

220 mechanism will be able to promote complete dominance of cooperation.

221 We note that none of these mechanisms allow coexistence of cooperation and
 222 defection. Therefore, we move on to the final example, applying SCT to micro-
 223 bial cooperation, where nonlinear benefits and environmental context can promote
 224 coexistence of cooperators and defectors.

225 Microbial cooperation and public goods

226 Here, we consider a simple, phenomenological model of competition between cooper-
 227 ating and cheating yeast in a cooperative sucrose-metabolism system, where invertase
 228 production by cooperators allows cheaters to utilize glucose, a public good released
 229 through sucrose hydrolysis (Carlson and Botstein, 1982; Dickinson, 1998; Gore et al.,
 230 2009). Specifically, invertase production by cooperators incurs a cost c and gener-
 231 ates a total benefit of 1 that is captured privately with efficiency ϵ . Assuming a
 232 linear relationship between glucose availability and microbial growth, the payoffs for
 233 cooperators π_C and cheaters (defectors) π_D can then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

234 where the use of glucose depends on the fraction of cooperators f . The dynamics of
 235 cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

236 For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of
 237 f (Figure 4A). Therefore, cooperation is favored when efficiency is greater than the
 238 cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than
 239 efficiency, $c > \epsilon$ (Gore et al., 2009).

240 Our framework allows us to reinterpret this result from the perspective of coexis-
 241 tence theory. In particular, the two strategies have no niche difference, corresponding
 242 to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by
 243 the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

244 Therefore, this model always predicts competitive exclusion, except in the neutral
 245 case $c = \epsilon$. The outcome is determined entirely by the fitness difference between
 246 cooperators and defectors, rather than by stabilizing niche differences. This example
 247 resembles the prisoner's dilemma discussed in the previous example (Figure 3).

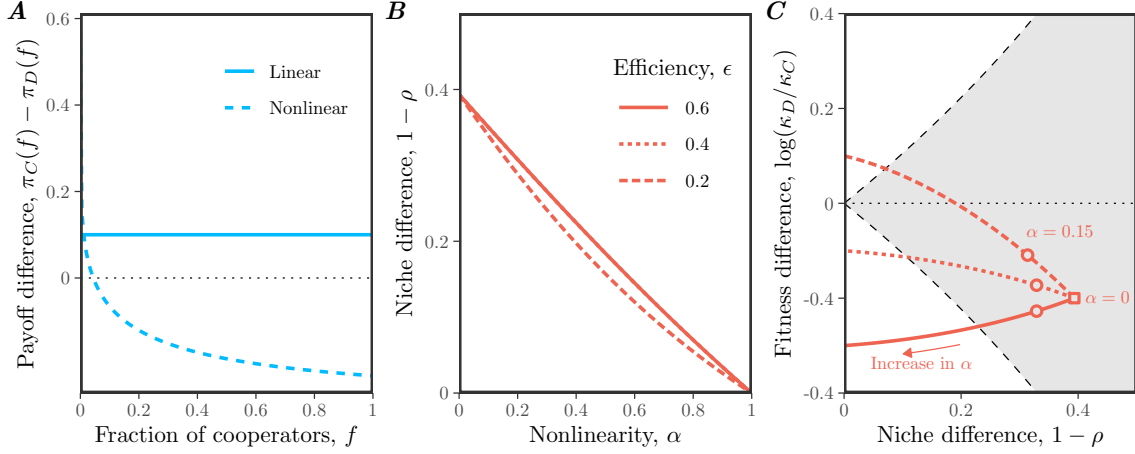


Figure 4: **Effects of nonlinear microbial growth on coexistence of cooperation and defection.** (A) Payoff differences for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.3$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . Note that lines for $\epsilon = 0.6$ and $\epsilon = 0.4$ are completely overlapping. (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha \rightarrow 0^+$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.3$, while allowing ϵ and α to vary.

248 In real biological systems, the effect of glucose on microbial growth can follow a
 249 nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

250 where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial
 251 growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are
 252 described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\{\epsilon + f(1 - \epsilon)\}^\alpha - c - \{f(1 - \epsilon)\}^\alpha]. \quad (19)$$

253 Under this nonlinear assumption, the payoff difference decreases with the cooperator
 254 fraction f , which can promote the coexistence of cooperators and defectors (Figure
 255 4A): cooperators have a competitive advantage when rare, whereas defectors have a
 256 competitive advantage when cooperators are abundant.

257 From a coexistence-theoretic perspective, this means that nonlinear growth can
 258 generate sufficient stabilizing niche difference to overcome the fitness difference. In

particular, the niche and fitness differences for this model correspond to:

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For example, the cost of cooperation does not affect niche difference and only affects fitness difference. Instead, niche difference is generated by the concave relationship between glucose availability and microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0 (Figure 4B,C). In contrast, as $\alpha \rightarrow 0^+$, the fitness difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

which no longer depends on the private capture efficiency ϵ (Figure 4C). Thus, strong concavity not only generates stabilizing niche differences, but also acts as an equalizing mechanism (Figure 4C).

Discussion

Standard evolutionary game theory and replicator equations provide powerful tools for predicting the outcome of strategy competition, including strategic dominance, coexistence, and bistability (Hofbauer and Sigmund, 1998; Cressman and Tao, 2014). However, these outcomes are typically understood by analyzing payoff differences within individual games, which makes it difficult to compare coexistence mechanisms across models. Building on Chesson's Modern Coexistence Theory from community ecology (Chesson, 2000), we developed a strategic coexistence theory (SCT) that allows us to decompose strategy competition in terms of stabilizing niche differences and fitness differences. This decomposition provides a unifying framework for comparing different games from the perspective of coexistence theory, allowing us to identify stabilizing and equalizing mechanisms for strategy competition.

To demonstrate the utility of SCT, we considered three case studies. First, the analysis of classic two-strategy games showed that SCT recovers the original classification for these classic games, highlighting the similarities between evolutionary games and community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025). Then, we compared different mechanisms for the evolution of cooperation, which showed that each mechanism promotes cooperation via different dynamical routes (Nowak, 2006). Kin selection, network reciprocity, and group selection primarily affect the fitness difference between cooperation and defection, whereas direct and indirect reciprocity can destabilize competition and generate priority effects. Finally, our analysis of microbial cooperation showed that coexistence between cooperators and defectors requires more than reducing the fitness cost

292 of cooperation: nonlinear benefits can generate stabilizing niche differences while
293 also reducing fitness asymmetry (Gore et al., 2009). Taken together, these examples
294 show that SCT can provide new insights into ecological underpinnings of strategy
295 competition.

296 There are several limitations to SCT. First, similar to MCT from community
297 ecology, SCT only allows for comparisons of pairwise strategies and therefore cannot
298 determine the outcomes of games with more than two strategies (Chesson, 2000).
299 Second, we focused on classic replicator equations, but Tarnita and Traulsen (2025)
300 recently pointed out that these classic formulations implicitly ignore differences in
301 intrinsic growth rates. In such cases, replicator equations may fail to predict the out-
302 come of competition (Tarnita and Traulsen, 2025), and a full Lotka-Volterra model is
303 needed to describe the abundances, rather than frequencies, of strategies. Thus, SCT
304 inherits the same assumptions as the classic replicator framework: it is most appro-
305 priate when strategy competition can be described in terms of relative frequencies
306 alone. When strategies differ intrinsically in growth, MCT may be required instead
307 to tease apart mechanisms of strategy coexistence. Finally, we have focused on simple
308 examples to illustrate the utility of SCT, but many games involve additional com-
309 plexities, including population structure (Nowak et al., 2010), stochasticity (Wang
310 et al., 2025; Bodin et al., 2026), and explicit ecological feedback. Extending SCT to
311 these settings will be necessary to determine how broadly stabilizing and equalizing
312 mechanisms can be compared across evolutionary games.

313 In conclusion, SCT provides a bridge between evolutionary game theory and
314 community ecology, allowing strategy competition to be understood from the per-
315 spective of coexistence theory. Therefore, rather than replacing existing EGT or
316 replicator-dynamics frameworks, SCT provides a complementary tool for teasing
317 apart the coexistence mechanisms underlying evolutionary games. More broadly,
318 our study builds on recent efforts to extend MCT beyond species competition (Park
319 et al., 2024, 2026), providing a foundation for building unifying coexistence theory
320 for competing entities, including species, pathogen strains, and behavioral strategies.

Supplementary Text

S1 Derivation of the strategic coexistence theory

To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

where x_i represents the frequency of strategy i , π_i represents the payoff of strategy i , and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

we obtain the following inequality that defines conditions for mutual invasion and long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

334 S2 Applications to classic two-strategy games

335 Consider a two-strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S11})$$

336 There are four classic games that we can define from this simple payoff matrix:
 337 prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt
 338 game ($R > T, P > S$), and harmony game ($R > T, S > P$). Here, we show that
 339 each game corresponds to a distinct region in the niche-fitness difference space under
 340 SCT.

341 To do so, we first begin by deriving expressions for niche overlap ρ and fitness
 342 difference κ_j/κ_i . Writing x_1 to denote the frequency of strategy 1, the expected
 343 payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S12})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S13})$$

344 Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S14})$$

$$W_1(0) = \exp(S) \quad (\text{S15})$$

$$W_2(1) = \exp(T) \quad (\text{S16})$$

$$W_2(0) = \exp(P) \quad (\text{S17})$$

345 Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \quad (\text{S18})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (\text{S19})$$

346 As before, SCT allows us to predict the mutual invasion of two strategy using the
 347 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S20})$$

348 Then, simple algebra reveals that each condition in mutual invasion inequality
 349 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S21})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S22})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S23})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S24})$$

350 These inequalities then allow us to classify outcomes of two-strategy games from the
 351 perspective of SCT. Note that the inequality between R and P does not affect the
 352 outcome.

353 **S3 Applications to mechanisms for the evolution of coopera-** 354 **tion**

355 We consider five mechanisms for the evolution of cooperation, all of which can be
 356 represented as a simple 2×2 payoff matrix (Nowak, 2006). Here, we go through each
 357 mechanism and derive measures for niche and fitness differences.

358 **Prisoner's dilemma.** Here, we consider a special case of Prisoner's dilemma
 359 which can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c \\ b & 0 \end{pmatrix}, \quad (\text{S25})$$

360 where b represents the benefit for the recipient and c represents cost for the donor.
 361 In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0 \quad (\text{S26})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c). \quad (\text{S27})$$

362 Therefore, as long as the cost is greater than zero, prisoner's dilemma always results
 363 in the competitive exclusion of cooperation.

364 **Kin selection.** Kin selection can be described by the following payoff matrix:

$$\begin{pmatrix} (b - c)(1 + r) & br - c \\ b - rc & 0 \end{pmatrix}, \quad (\text{S28})$$

365 where r represents the genetic relatedness. In this case, the niche and fitness differ-
 366 ences correspond to:

$$1 - \rho = 0 \quad (\text{S29})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c - br). \quad (\text{S30})$$

367 Therefore, defection will exclude cooperation when $c > br$. Equivalently, cooperation
 368 will be favored when $r > c/b$, which recovers the classic result.

369 **Direct reciprocity.** Direct reciprocity can be described by the following payoff
 370 matrix:

$$\begin{pmatrix} (b - c)/(1 - w) & -c \\ b & 0 \end{pmatrix}, \quad (\text{S31})$$

where w represents the probability of next round. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)/(1 - w) + c - b}{2}\right) \quad (\text{S32})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{b + c - (b - c)/(1 - w)}{2}\right). \quad (\text{S33})$$

Therefore, defection will exclude cooperation when

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S34})$$

$$\iff \exp\left(\frac{(b - c)/(1 - w) + c - b}{2}\right) < \exp\left(\frac{b + c - (b - c)/(1 - w)}{2}\right) \quad (\text{S35})$$

$$\iff bw < c \quad (\text{S36})$$

However, note that the following equality holds whenever $c > 0$:

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}. \quad (\text{S37})$$

Therefore, when $w > c/b$, we will have

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1} < \rho, \quad (\text{S38})$$

which implies priority effect (equivalently, bistability). Even as $w \rightarrow 1^-$, this inequality holds and therefore, cooperation cannot competitively exclude defection.

Indirect reciprocity. Indirect reciprocity can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c(1 - q) \\ b(1 - q) & 0 \end{pmatrix}, \quad (\text{S39})$$

where q represents social acquaintanceship. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)q}{2}\right), \quad (\text{S40})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{2c - (b + c)q}{2}\right). \quad (\text{S41})$$

In this case, the condition for the dominance of defection (and therefore competitive exclusion of cooperation) can be written as:

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S42})$$

$$\iff bq < c \quad (\text{S43})$$

384 Moreover, the following inequality always holds as long as $c > 0$.

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}, \quad (\text{S44})$$

385 which implies that coexistence between cooperation and defection is not possible.
 386 Therefore, when $q < c/b$, then defection will competitively exclude cooperation.
 387 Otherwise, the system will enter the priority-effect regime.

388 Interestingly, when $q \rightarrow 1$, we have

$$\rho = \exp\left(\frac{b-c}{2}\right) \quad (\text{S45})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{c-b}{2}\right) \quad (\text{S46})$$

$$\frac{1}{\rho} = \exp\left(\frac{c-b}{2}\right) \quad (\text{S47})$$

$$(\text{S48})$$

389 In other words, the following inequality will hold:

$$\frac{1}{\rho} = \frac{\kappa_2}{\kappa_1} < \rho \quad (\text{S49})$$

390 **Network reciprocity.** Network reciprocity can be described by the following
 391 payoff matrix:

$$\begin{pmatrix} b-c & H-c \\ b-H & 0 \end{pmatrix}, \quad (\text{S50})$$

392 where $H = [(b-c)k - 2c]/[(k+1)(k-2)]$ and k represents the number of neighbors.
 393 In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S51})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(-H + c), \quad (\text{S52})$$

394 which implies that there is complete niche overlap and that fitness difference decreases
 395 with H . Since H is a monotonically decreasing function of k for $k > 2$, fitness
 396 difference will increase with k . In particular, cooperation will competitively exclude
 397 defection when $H > c$, meaning that:

$$b/c > k. \quad (\text{S53})$$

398 **Group selection.** Group selection can be described by the following payoff
 399 matrix:

$$\begin{pmatrix} (b-c)(m+n) & (b-c)m - cn \\ bn & 0 \end{pmatrix}, \quad (\text{S54})$$

where m represents the number of groups and n represents the group size. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S55})$$

$$\frac{\kappa_2}{\kappa_1} = \exp((c - b)m + cn), \quad (\text{S56})$$

which implies that there is complete niche overlap. The fitness difference will increase with n but decrease with m , provided that $b > c$. In particular, cooperation will competitively exclude defection under the following condition:

$$1 + \frac{n}{m} < \frac{b}{c}. \quad (\text{S57})$$

References

- Adler, P. B., M. Detto, S. P. Ellner, T. L. Gibbs, Z. J. Gold, S. J. Leonard, J. I. Levine, A. E. Schiffer, C. Song, M. Stemkovski, et al. (2026). What have we learned from empirical applications of modern coexistence theory? *Ecology letters* 29(6), e70404.
- Allesina, S. (2026). Global stability of ecological and evolutionary dynamics via equivalence. *Proceedings of the National Academy of Sciences* 123(13), e2534915123.
- Archetti, M. and I. Scheuring (2011). Coexistence of cooperation and defection in public goods games. *Evolution* 65(4), 1140–1148.
- Axelrod, R. and W. D. Hamilton (1981). The evolution of cooperation. *science* 211(4489), 1390–1396.
- Bodin, F., G. Wang, and J. B. Plotkin (2026). Eco-evolutionary games in noisy environments. *bioRxiv*, 2026–05.
- Carlson, M. and D. Botstein (1982). Two differentially regulated mRNAs with different 5' ends encode secreted and intracellular forms of yeast invertase. *Cell* 28(1), 145–154.
- Chesson, P. (2000). Mechanisms of maintenance of species diversity. *Annual review of Ecology and Systematics* 31(1), 343–366.
- Cooney, D. B. (2020). Analysis of multilevel replicator dynamics for general two-strategy social dilemma. *Bulletin of Mathematical Biology* 82(6), 66.
- Cordero, O. X., L.-A. Ventouras, E. F. DeLong, and M. F. Polz (2012). Public good dynamics drive evolution of iron acquisition strategies in natural bacterioplankton populations. *Proceedings of the National Academy of Sciences* 109(49), 20059–20064.

430 Cressman, R. (2003). *Evolutionary dynamics and extensive form games*, Volume 5.
431 MIT Press.

432 Cressman, R. and Y. Tao (2014). The replicator equation and other game dynamics.
433 *Proceedings of the National Academy of Sciences* 111(supplement_3), 10810–10817.

434 Dickinson, J. R. (1998). *Metabolism and molecular physiology of Saccharomyces*
435 *cerevisiae*. Taylor & Francis.

436 Doebeli, M. and C. Hauert (2005). Models of cooperation based on the Prisoner’s
437 Dilemma and the Snowdrift game. *Ecology letters* 8(7), 748–766.

438 Doebeli, M., C. Hauert, and T. Killingback (2004). The evolutionary origin of coop-
439 erators and defectors. *science* 306(5697), 859–862.

440 Godoy, O., N. J. Kraft, and J. M. Levine (2014). Phylogenetic relatedness and the
441 determinants of competitive outcomes. *Ecology letters* 17(7), 836–844.

442 Godoy, O. and J. M. Levine (2014). Phenology effects on invasion success: insights
443 from coupling field experiments to coexistence theory. *Ecology* 95(3), 726–736.

444 Gokhale, C. S. and A. Traulsen (2025). Higher-order equivalence of Lotka-Volterra
445 and replicator dynamics. *bioRxiv*, 2025–03.

446 Gore, J., H. Youk, and A. Van Oudenaarden (2009). Snowdrift game dynamics and
447 facultative cheating in yeast. *Nature* 459(7244), 253–256.

448 Hamilton, W. D. (1964). The genetical evolution of social behaviour. II. *Journal of*
449 *theoretical biology* 7(1), 17–52.

450 Hauert, C. (2002). Effects of space in 2×2 games. *International Journal of Bifur-*
451 *cation and Chaos* 12(07), 1531–1548.

452 Hauert, C. and M. Doebeli (2004). Spatial structure often inhibits the evolution of
453 cooperation in the snowdrift game. *Nature* 428(6983), 643–646.

454 Hofbauer, J. (1981). On the occurrence of limit cycles in the volterra-lotka equation.
455 *Nonlinear Analysis: Theory, Methods & Applications* 5(9), 1003–1007.

456 Hofbauer, J. and K. Sigmund (1998). *Evolutionary games and population dynamics*.
457 Cambridge university press.

458 Kocher, S. D., R. Mallarino, B. E. Rubin, D. W. Yu, H. E. Hoekstra, and N. E.
459 Pierce (2018). The genetic basis of a social polymorphism in halictid bees. *Nature*
460 *Communications* 9(1), 4338.

461 Kraft, N. J., O. Godoy, and J. M. Levine (2015). Plant functional traits and the mul-
462 tidimensional nature of species coexistence. *Proceedings of the National Academy*
463 *of Sciences* 112(3), 797–802.

464 Lewontin, R. C. (1961). Evolution and the theory of games. *Journal of Theoretical*
465 *Biology* 1(3), 382–403.

466 Lieberman, E., C. Hauert, and M. A. Nowak (2005). Evolutionary dynamics on
467 graphs. *Nature* 433(7023), 312–316.

468 Lin, H., D. Wang, Q. Wang, J. Mao, Y. Bai, and J. Qu (2024). Interspecific compe-
469 tition prevents the proliferation of social cheaters in an unstructured environment.
470 *The ISME Journal* 18(1), wrad038.

471 May, R. M. (1987). More evolution of cooperation. *Nature* 327(6117), 15–17.

472 Motro, U. (1991). Co-operation and defection: playing the field and the ESS. *Journal*
473 *of Theoretical Biology* 151(2), 145–154.

474 Nowak, M. A. (2006). Five rules for the evolution of cooperation. *science* 314(5805),
475 1560–1563.

476 Nowak, M. A. and R. M. May (1992). Evolutionary games and spatial chaos. *na-*
477 *ture* 359(6398), 826–829.

478 Nowak, M. A. and K. Sigmund (1998). Evolution of indirect reciprocity by image
479 scoring. *Nature* 393(6685), 573–577.

480 Nowak, M. A. and K. Sigmund (2005). Evolution of indirect reciprocity. *Na-*
481 *ture* 437(7063), 1291–1298.

482 Nowak, M. A., C. E. Tarnita, and T. Antal (2010). Evolutionary dynamics in struc-
483 tured populations. *Philosophical Transactions of the Royal Society B: Biological*
484 *Sciences* 365(1537), 19.

485 Page, K. M. and M. A. Nowak (2002). Unifying evolutionary dynamics. *Journal of*
486 *theoretical biology* 219(1), 93–98.

487 Park, S. W., S. Cobey, C. J. E. Metcalf, J. M. Levine, and B. T. Grenfell (2024).
488 Predicting pathogen mutual invasibility and co-circulation. *Science* 386(6718),
489 175–179.

490 Park, S. W., J. Levine, and B. Grenfell (2026). Unifying coexistence theory for
491 ecological communities and pathogen strain competition. *bioRxiv*, 2026–05.

492 Pollak, S., M. Gralka, Y. Sato, J. Schwartzman, L. Lu, and O. X. Cordero (2021).
493 Public good exploitation in natural bacterioplankton communities. *Science Ad-*
494 *vances* 7(31), eabi4717.

495 Schuster, P. and K. Sigmund (1983). Replicator dynamics. *Journal of theoretical*
496 *biology* 100(3), 533–538.

Shalizi, C. R. (2009). Dynamics of bayesian updating with dependent data and misspecified models. *arXiv*.

Sieben, A. J., J. R. Mihaljevic, and L. G. Shoemaker (2022). Quantifying mechanisms of coexistence in disease ecology. *Ecology* 103(12), e3819.

Slobodkin, L. B. (1964). The strategy of evolution. *American scientist* 52(3), 342–357.

Smith, J. M. and G. R. Price (1973). The logic of animal conflict. *Nature* 246(5427), 15–18.

Souza, M. O., J. M. Pacheco, and F. C. Santos (2009). Evolution of cooperation under N-person snowdrift games. *Journal of Theoretical Biology* 260(4), 581–588.

Steidinger, B. S. and J. D. Bever (2014). The coexistence of hosts with different abilities to discriminate against cheater partners: an evolutionary game-theory approach. *The American Naturalist* 183(6), 762–770.

Szabó, G. and G. Fath (2007). Evolutionary games on graphs. *Physics reports* 446(4–6), 97–216.

Tarnita, C. E. and A. Traulsen (2025). Reconciling ecology and evolutionary game theory or “When not to think cooperation”. *Proceedings of the National Academy of Sciences* 122(14), e2413847122.

Taylor, P. D. and L. B. Jonker (1978). Evolutionary stable strategies and game dynamics. *Mathematical biosciences* 40(1–2), 145–156.

Traulsen, A. and M. A. Nowak (2006). Evolution of cooperation by multilevel selection. *Proceedings of the National Academy of Sciences* 103(29), 10952–10955.

Trivers, R. L. (1971). The evolution of reciprocal altruism. *The Quarterly review of biology* 46(1), 35–57.

Wang, G., Q. Su, L. Wang, and J. B. Plotkin (2025). Evolution of social behaviors in noisy environments. *arXiv*.

Wedekind, C. and M. Milinski (2000). Cooperation through image scoring in humans. *Science* 288(5467), 850–852.

West, S. A., A. S. Griffin, and A. Gardner (2007). Evolutionary explanations for cooperation. *Current biology* 17(16), R661–R672.